

3.7 Tangent Planes and Normal Vectors Worksheet

1. Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $(1, 2, 3)$.

To create the equation of a plane we need a point on the plane and a vector normal to the plane.

Let $F(x, y, z) = x^2 + y^2 + z^2$. We know that $\vec{\nabla}F(1, 2, 3)$ will represent a normal vector to the plane.

We calculate $\vec{\nabla}F = \langle 2x, 2y, 2z \rangle \rightarrow \vec{\nabla}F(1, 2, 3) = \langle 2, 4, 6 \rangle$. We also know that $(1, 2, 3)$ is a point in the plane.

Therefore, the equation of the tangent plane is given by:

$$2(x-1) + 4(y-2) + 6(z-3) = 0$$

2. Find the equation of the tangent plane at the point $(0, 9, 4)$ to the surface $z = e^x \sqrt{y} + x^2 + 1$.

To create the equation of a plane we need a point on the plane and a vector normal to the plane.

We will rewrite our given equation as $z = e^x \sqrt{y} + x^2 + 1 \rightarrow z - e^x \sqrt{y} - x^2 - 1 = 0$ and then let

$F(x, y, z) = z - e^x \sqrt{y} - x^2 - 1$. We know that $\vec{\nabla}F(0, 9, 4)$ will represent a normal vector to the plane.

We calculate $\vec{\nabla}F = \left\langle -e^x \sqrt{y} - 2x, -\frac{1}{2}e^x y^{-1/2}, 1 \right\rangle \rightarrow \vec{\nabla}F(0, 9, 4) = \left\langle -3, -\frac{1}{6}, 1 \right\rangle$. We also know that $(0, 9, 4)$ is a point in the plane.

Therefore, the equation of the tangent plane is given by:

$$-3(x-0) - \frac{1}{6}(y-9) + 1(z-4) = 0$$

3. Let $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$

a) Find the equation of the tangent plane at the point $(1,1,1)$ to the surface $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$.

To create the equation of a plane we need a point on the plane and a vector normal to the plane.

Let $F(x, y, z) = x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30}$. We know that $\vec{\nabla}F(1,1,1)$ will represent a normal vector to the plane.

We calculate

$$\vec{\nabla}F = \langle 10x^9 + 10x^9y^{10} + 10x^9y^{20}z^{30}, 10y^9x^{10} + 20y^{19}x^{10}z^{30}, 30z^{29}x^{10}y^{20} \rangle \rightarrow \vec{\nabla}F(1,1,1) = \langle 30, 30, 30 \rangle.$$

We also know that $(1,1,1)$ is a point in the plane.

Therefore, the equation of the tangent plane is given by:

$$30(x-1) + 30(y-1) + 30(z-1) = 0$$

b) Determine the linear approximation of the function $x^{10} + x^{10}y^{10} + x^{10}y^{20}z^{30} = 3$ at $(1,1,1)$ and use it to approximate the value of the z -coordinate of the surface at the point $(0.8, 0.9)$.

The equation of the tangent plane from part (a) was

$$30(x-1) + 30(y-1) + 30(z-1) = 0 \rightarrow (x-1) + (y-1) + (z-1) = 0 \rightarrow z = 3 - x - y$$

Therefore, a linearization of the surface at $(1,1,1)$ would be $L(x, y) = 3 - x - y$ and we can use this function to approximate values on the surface which are near $(1,1)$.

$$\text{Thus, } z\text{-value at } (0.8, 0.9) \approx L(0.8, 0.9) = 3 - (0.8) - (0.9) = 3 - 1.7 = 1.3$$

4. Let $f(x, y) = 2y \sin(x - y)$.

a) Find an equation of the tangent plane to $f(x, y)$ at the point $(1, 1, 0)$.

To create the equation of a plane we need a point on the plane and a vector normal to the plane.

We will rewrite our given equation as $z = 2y \sin(x - y) \rightarrow z - 2y \sin(x - y) = 0$ and then let

$F(x, y, z) = z - 2y \sin(x - y)$. We know that $\vec{\nabla} F(1, 1, 0)$ will represent a normal vector to the plane.

We calculate $\vec{\nabla} F = \langle -2y \cos(x - y), -2 \sin(x - y) + 2y \cos(x - y), 1 \rangle \rightarrow \vec{\nabla} F(1, 1, 0) = \langle -2, 2, 1 \rangle$. We also know that $(1, 1, 0)$ is a point in the plane.

Therefore, the equation of the tangent plane is given by:

$$-2(x - 1) + 2(y - 1) + 1(z - 0) = 0$$

b) Determine the linear approximation of the function $f(x, y)$ at $(1, 1)$ and use it to approximate $f(1.2, 0.9)$.

The equation of the tangent plane from part (a) was

$$-2(x - 1) + 2(y - 1) + 1(z - 0) = 0 \rightarrow z = 2(x - 1) - 2(y - 1) \rightarrow z = 2x - 2y$$

Therefore, a linearization of $f(x, y)$ at $(1, 1)$ would be $L(x, y) = 2x - 2y$ and we can use this function to approximate values on the surface which are near $(1, 1)$.

$$\text{Thus, } f(1.2, 0.9) \approx L(1.2, 0.9) = 2(1.2) - 2(0.9) = 2.4 - 1.8 = 0.6$$